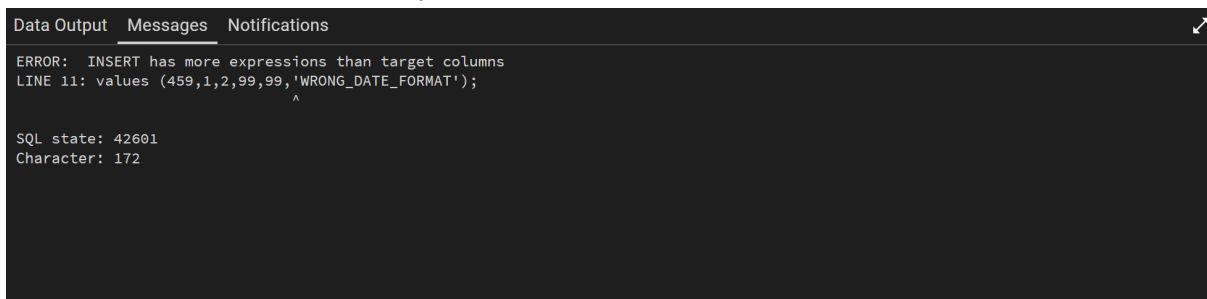


1. Lab 01 – Testing Transaction Atomicity

```
begin;  
update rental  
set return_date = NOW() where rental_id = 2;  
insert into payment(  
    customer_id,  
    staff_id,  
    rental_id,  
    amount,  
    payment_date  
)  
values (459,1,2,99,99,'WRONG_DATE_FORMAT');  
commit;
```

The insert statement intentionally uses an invalid value:



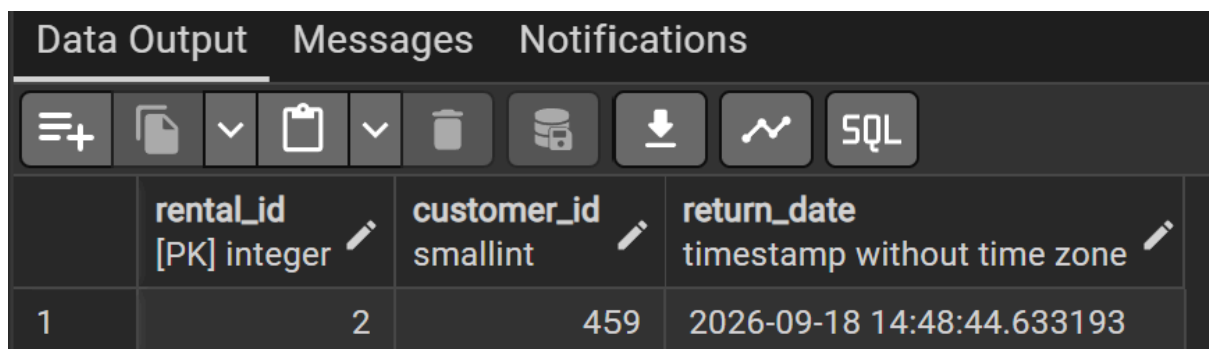
```
ERROR: INSERT has more expressions than target columns  
LINE 11: values (459,1,2,99,99,'WRONG_DATE_FORMAT');  
                                ^  
  
SQL state: 42601  
Character: 172
```

This causes the transaction to fail.

After the error, the transaction was rolled back: ROLLBACK;

Then the rental record was checked:

```
SELECT rental_id, customer_id, return_date FROM rental WHERE rental_id = 2;
```



	rental_id [PK] integer	customer_id smallint	return_date timestamp without time zone
1	2	459	2026-09-18 14:48:44.633193

The return_date remained unchanged after the rollback.

This shows that although the update statement was executed before the error, its changes were not permanently saved.

This exercise demonstrates **Atomicity** in ACID. When one operation in the transaction fails, the entire transaction is rolled back. Therefore, the database remains in its previous consistent state.

Exercise 2:

Create Staff and SAVEPOINT

```
begin;
insert into staff (
    first_name,
    last_name,
    address_id,
    email,
    store_id,
    active,
    username,
    password
)
values ('Hoang','Hai Van',1,'11247244@st.neu',1,True,'fda_student_ds66','11247244');

savepoint staff_created;
```

Data Output	Messages	Notifications
SAVEPOINT		
Query returned successfully in 73 msec.		

Perform an Invalid Operation

Data Output	Messages	Notifications
ERROR: insert or update on table "staff" violates foreign key constraint "staff_store_id_fkey" Key (store_id)=(999) is not present in table "store".		
SQL state: 23503 Detail: Key (store_id)=(999) is not present in table "store".		

ROLLBACK TO staff_created;

Then

UPDATE staff SET store_id = 1 WHERE username = 'fda_student_ds66'; COMMIT;

Data Output Messages Notifications					
≡+ 📄 ▼ 📋 ▼ 🗑️ 🗄️ ⬇️ 📈 SQL Showing rows: 1 to 5 ✎ 🖨️					
	staff_id [PK] integer ✎	first_name character varying (45) ✎	last_name character varying (45) ✎	username character varying (16) ✎	store_id smallint ✎
1	3	Hoang	Hai Van	fda_student_ds66	1
2	4	Hoang	Hai Van	fda_student_ds66	1
3	5	Hoang	Hai Van	fda_student_ds66	1
4	6	Hoang	Hai Van	fda_student_ds66	1
5	7	Hoang	Hai Van	fda_student_ds66	1